

Prio

Quiet and compact mixed-flow fan

- Available with AC and EC motors for 50 and 60Hz
- Easy to install
- Sound and energy optimized impellers

[Find more details in our online catalogue](#)



Flexibility

The prio fans are intended to use for **supply** or **extract** air, designed to be installed in **any position**. This ensures that the fans can be used in a variety of **commercial** and **domestic** applications.

Reliability

The unit's **airtight casing** and corrosion resistance allows to use indoors and in damp areas. The combination of a reliable casing and motor construction ensures to minimize the need for maintenance of the fans and allows long **continuous operation**.

Performance

The **sound optimized** impellers together with **high efficient** external rotor motors are designed to ensure high-level performance to **minimize power consumption** and **maximize efficiency**.

Accessories

prio fan can be selected together with variety types of **accessories** as fast clamps, heaters, filters, dampers, etc.

Certifications



Ecodesign (ErP) compliant

Compliant with today's and tomorrow's European Union rules and requirements for energy labelling and ecodesign for Energy-related Products (ErP).



Eco Platform: EPD Verified

This product comes with an Environmental Product Declaration (EPD), published by EPD Norway, a global EPD program operator and member of the ECO Platform (<https://www.eco-platform.org/who-is-participating.html>). Compliant with ISO 14025 and EN 15804, Systemair EPDs support Life Cycle Assessments (LCAs) in all markets where the product is used. Scroll down to download the EPDs.

Features

Construction

Casing made out of special composite material or sheet metal, is fully airtight class C according to EN12237.
The Fans are equipped with an external **terminal box with** protection class **IP44**.

Impeller

prio fans have aerodynamically optimized impellers and guide vanes. These are from lightweight, hard-wearing fiberglass reinforced **polyamide**, dynamically **balanced** and paired with corresponding external rotor motors.

Motor

Depending on the type, prio fans are equipped with **AC** or **EC** external rotor motor. Motors are suitable for **50Hz** and **60Hz**.

Motor protection

Fans with **AC** motors have an **integrated** thermal protection with manual (electrical) reset.

For **size 315 – 500** an external motor protection is needed.

Fans with **EC** motors have an **integrated** electronic **thermal protection** including **locked-rotor protection** and **soft-start**.

Control

EC motors with **built-in potentiometer** to adjust working point can be also controlled by external **signal 0-10V**.

AC motors can be controlled by **5-step** or **stepless** speed regulator.

Installation

prio fans can be installed in any position **indoor** and in **damp areas**. Spigot with 25 mm length is perfectly designed to connect circular duct by **fast clamps preventing vibration** to the duct.

Technical parameters

Nominal data

Voltage (nominal)	230	V
Frequency	60	Hz
Phases	1~	
Input power	288	W
Input power kW	0.288	kW
Input current	1.256	A
Impeller speed	2,929	rpm
Air flow	max 2,282	m³/h
Specific ratio	1,000000	
Capacitance of capacitor	6	µF
Temperature of transported air	max 55	°C
Max temperature of transported air, when speed controlled	55	°C

Protection/Classification

Enclosure class, motor	IP44
Insulation class	F

Data according to ErP

ErP ready	Not ErP relevant
Measurement category	D

Dimensions and weights

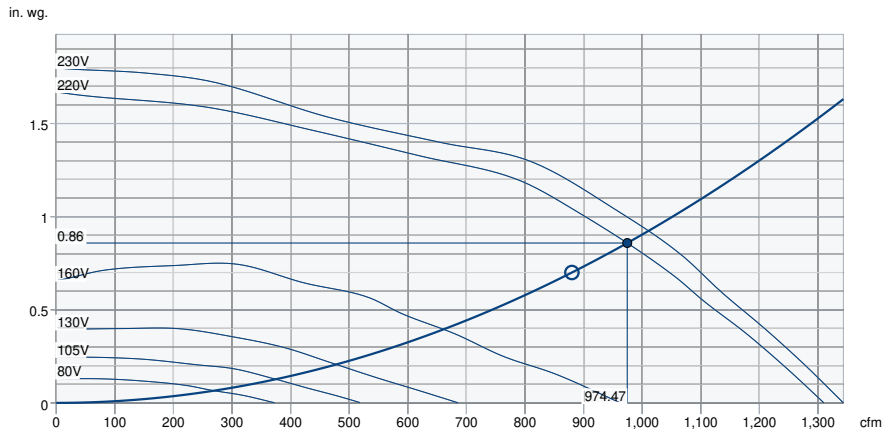
Duct dimension; Circular, inlet	250	mm
Duct dimension; Circular, outlet	250	mm
Weight	5.6	kg

Others

Duct connection type	Circular
Motor type	AC

Performance

Performance curve



Hydraulic data

Required air flow	880.00 cfm
Required static pressure	0.70 in. wg.
Working air flow	974.47 cfm
Working static pressure	0.86 in. wg.
Air density	1.204 kg/m³
Power	277.2 W
Fan control - RPM	2,814 rpm
Current	1.27 A
SFP	0.603 kW/m³/s
Control voltage	220.0 V
Supply voltage	220 V

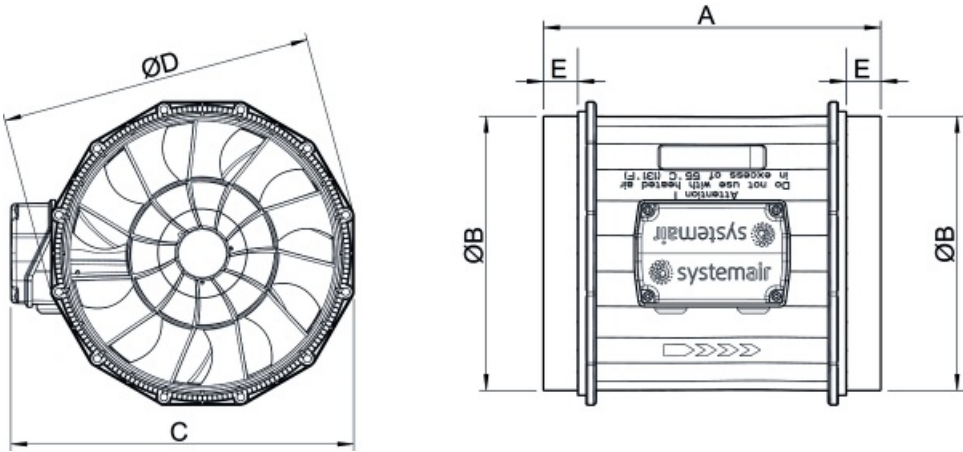
Sound power level		63	125	250	500	1k	2k	4k	8k	Total
Inlet	dB(A)	47	60	66	68	73	72	64	55	77
Outlet	dB(A)	50	61	67	70	74	72	64	55	78
Surrounding	dB(A)	17	38	48	52	58	56	43	28	61
Sound pressure level at 3m (20m² Sabine)	dB(A)	-	-	-	-	-	-	-	-	54
Sound pressure level at 3m free field	dB(A)	-	-	-	-	-	-	-	-	40

Ecodesign

Ecodesign 327

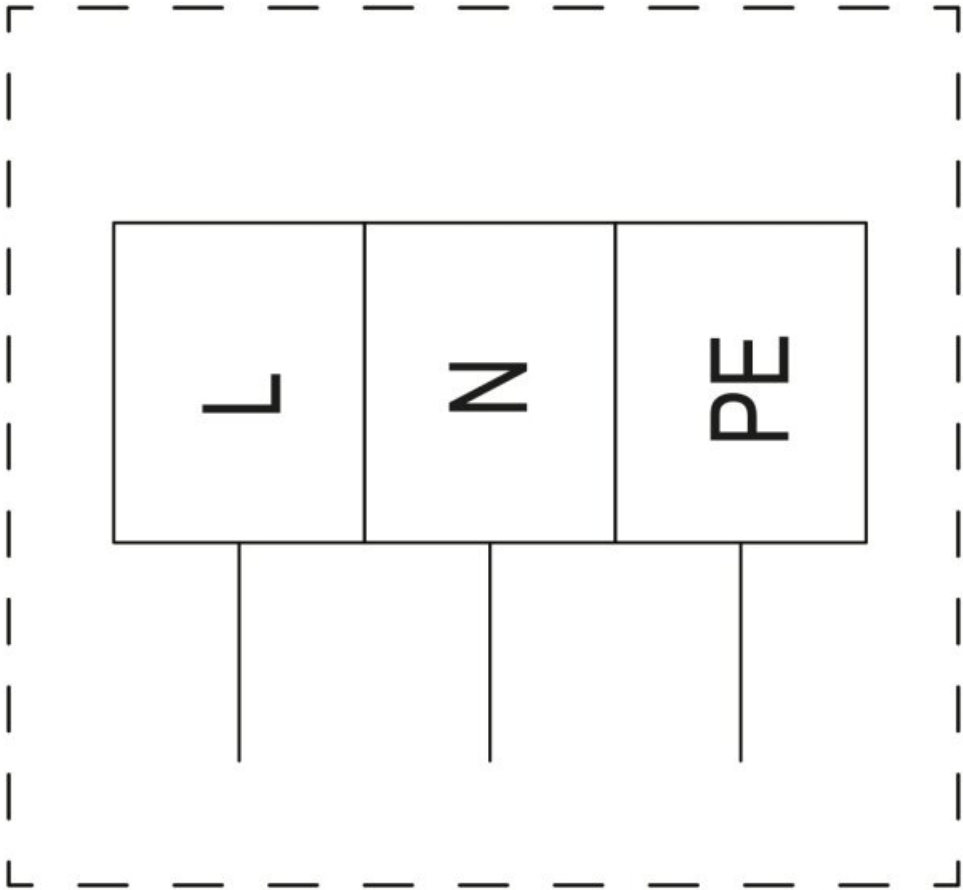
Manufacturer	Systemair GmbH		
Type	prio 250E2		
Year of manufacture	See name plate of the fan		
Air flow qv	1,400	m³/h	
Efficiency category	total		
Efficiency grade N	68.4		
Efficiency grade target N	58		
Speed (rpm) n	2,660	rpm	
Pressure increase total psf	277	Pa	
Power consumption Ped	190	W	
Overall efficiency	57.5	%	
Overall efficiency target	47.1	%	
Variable speed drive	No		
Additional components	Components used to calculate the energy efficiency that are not apparent from the measurement category are detailed in the CE declaration.		
Maintenance	Information on installation, operation and maintenance is provided in the operating instructions.		
Recycling / disposal	Information on recycling and disposal is provided in the operating instructions.		

Dimension



	A	øB	C	øD	E
prio 250	300	249	303	284	30

Wiring



Function / assignment

L	Power supply 230V AC, 50...60Hz
N	Neutral conductor
PE	Protective conductor
Speed control:	Voltage controllable (See accessories in ouronline catalogue)
Motor protection:	By integratet thermal contacts

Accessories

- Frequency converter FRQ5SE-6A (37421)
- RE 1,5 Speed control (5000)
- REPT 6 Digital regulator (5698)
- REU 1.5 Speed control (5004)
- SG 250 Protection guard (5610)
- CB 250-6,0 400V/2 Duct heater (5372)
- Frequency converter FRQSE-6A (37419)
- Presence detector/IR24-P (6995)
- RT 0-30 Room Thermostat (5151)
- CB 250-3,0 230V/1 Duct heater (5385)
- CBM 250-6,0 400V/2 Duct heater (9089)
- CBMF 250-6,0 400V/2 Duct heater (12249)
- FFR 250 Filter cassette (1776)
- FK 250 Fast clamp (1612)
- RSK-250 Back draft damper (5603)
- VBC 250-3 Water heating batt (9843)
- VK-25 Louvre shutter (5640)
- IGC-250 Intake grille (76878)
- IGC-LI-250 intake grill (92763)
- REE 2 Speed control (5316)
- RETP 6 Temp/Pressure regulator (32293)
- REV-3POL/03-7,5kW R/Y (33978)
- VKK-250 Back draft damper (1627)
- DTV500A (96807)
- HR1 Room Humidistat (215150)
- prio 250 mount. bracket compl. (79316)
- T 120 Timer (5165)
- CB 250-9,0 400V/3 Duct heater (5373)
- CBMF 250-3,0 230V/1 Duct heater (12248)
- CWK 250-3-2,5 Duct cooler, circ (30024)
- FGR 250 Filter cassette (1815)
- LDC 250-900 Silencer (5196)
- VBC 250-2 Water heating batt (5460)
- VBF 250 Water heating battery (1733)
- VK-25 Louvre shutter (87687)
- Room hygrostat HR-S (286251)

Documents

- imo_prioair_en_001
- prioair250_o_rahmen.dxf
- EU DECLARATION OF CONFORMITY_PRIO_PRIO-XP_EN_007.PDF